
CLINTON ENWEREM

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PROFESSIONAL SUMMARY

Ph.D. Candidate and Robotics Research Engineer with **4+ years** of experience in:

- Dexterous grasping and manipulation under multimodal uncertainty: grasp planning, safety-critical control (control barrier functions, safety filters), safe and risk-sensitive reinforcement learning, and imitation/behavior cloning, validated sim-to-real on real hardware (xArm7, FR3 + LEAP Hand, RealHand L6).
- Research-grade ROS 2 robot software development in Python and C++: URDF kinematics toolkits, hand grasp planning libraries, and simulation benchmarking in MuJoCo and Drake.

EDUCATION

University of Maryland, College Park

Ph.D., Electrical & Computer Engineering (Dexterous Manipulation). Expected Fall 2026.

MD, USA

Aug. 2021 – Present

University of Nigeria, Nsukka

B.Eng., Electrical Engineering (Control Theory Emphasis). Highest Honors. GPA: 3.84/4.0 (WES U.S. equivalent).

Enugu, Nigeria

Aug. 2018

WORK EXPERIENCE

Graduate Research Assistant

Institute for Systems Research (ISR), University of Maryland

Aug. 2021 – Present

College Park, MD

- Develop and maintain Python and C++ software for grasp planning, stochastic motion planning, and closed-loop manipulation control on real manipulators (xArm7, UR5, LEAP Hand).
- Build expert data collection pipelines and train MLP, LSTM, and DP3 behavior cloning policies for grasp stabilization under partial tactile observability.

Robotics Trainee

Robotics & Artificial Intelligence Nigeria

Mar. 2020 – Feb. 2021

Ibadan, Nigeria

- Built ROS-based visual-SLAM, state-estimation, and real-time flight-control software for a low-cost UAV, integrating onboard perception and control on embedded hardware.

SELECTED PROJECTS

Lightweight URDF Kinematics Toolkit (Code) | Python

Python library for URDF parsing and kinematics extraction: FK/IK, condition-number singularity detection, Damped Least Squares Jacobian resolution, and kinematics-aware C-space sampling.

Sep. – Oct. 2025

Collision-Free Parallel Hand IK Solver (Demo) | Python, Pinocchio, FCL, MeshCat

Damped Least Squares IK solver for a branched-chain robotic hand using per-finger Jacobians, real-time FCL collision checking, and MeshCat visualization; converges in 720 iterations to sub-0.1 mm error.

Aug. 2025

TECHNICAL SKILLS

Robotics: ROS/ROS2, MuJoCo, Drake, RViz, MoveIt2; **Control & Safe Learning:** Safe & Constrained Reinforcement Learning, Control Barrier Functions, Safety Filters, Constrained MDPs, Model Predictive Control (MPC), QP-based Control, Lyapunov Stability, Sim-to-Real Transfer; **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Programming:** Python, C++, MATLAB, Bash, $\LaTeX$; **Optimization:** Gurobi, Pyomo, Mosek, CVXPY; **RL Sandboxes:** OpenAI Gym, Safety Gymnasium, PyBullet; **Dev Tools:** git, GitHub, GitLab, Docker; **Robots & EoATs:** UR3e/UR5, xArm7, LEAP Hand, RealHand L6.

SELECTED PUBLICATIONS

- **C. Enwerem**, J. S. Baras, C. Belta. “EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows.” *Under review*, 2026. [arXiv:2606.12728](https://arxiv.org/abs/2606.12728).
- **C. Enwerem**, S. Kalyanaraman, J. S. Baras, C. Belta. “Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty.” *To appear, IROS 2026*. [arXiv:2604.25897](https://arxiv.org/abs/2604.25897).
- **C. Enwerem**, J. S. Baras, C. Belta. “Risk-Constrained Belief-Space Optimization for Safe Control under Latent Uncertainty.” *To appear, CDC 2026*. [arXiv:2604.03868](https://arxiv.org/abs/2604.03868).
- **C. Enwerem**, A. G. Puranic, J. S. Baras, C. Belta. “Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression.” *CDC 2025*. [arXiv:2506.06954](https://arxiv.org/abs/2506.06954).